

Two Body Oscillations



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Whatever we will discuss

- ✓ Two body oscillations
- ✓ Related mathematical problems

Two Body Oscillations

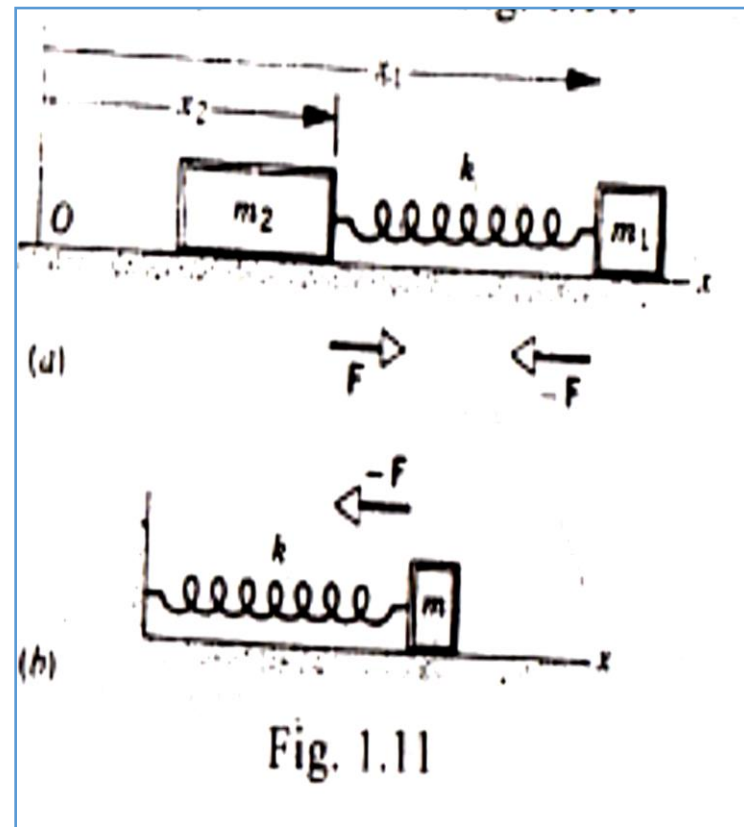
If displaced a small distance from equilibrium position, the molecule will oscillate about the equilibrium position.

➤ The relative separation $x_1 - x_2$ gives the length of the spring at any time.

➤ Unstretched length of the spring is L

➤ The change in length of the spring is

$$x = (x_1 - x_2) - L.$$

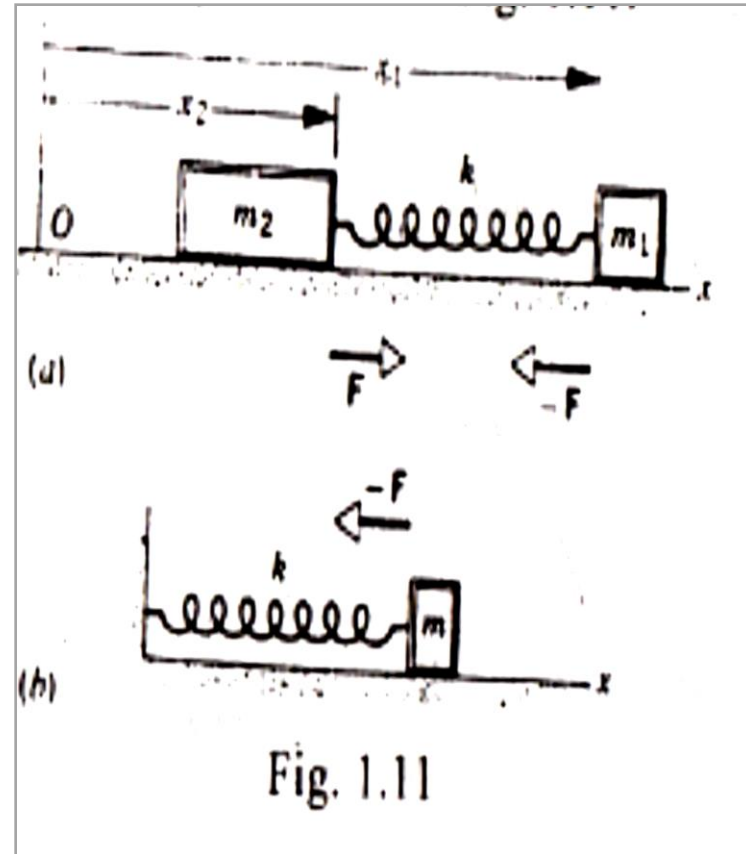


Two Body Oscillations

- The magnitude of the force that the spring exerts on each particle is

$$F = kx.$$

- If the spring exerts a force $-F$ on m_1 , then it exerts a force $+F$ on m_2 .

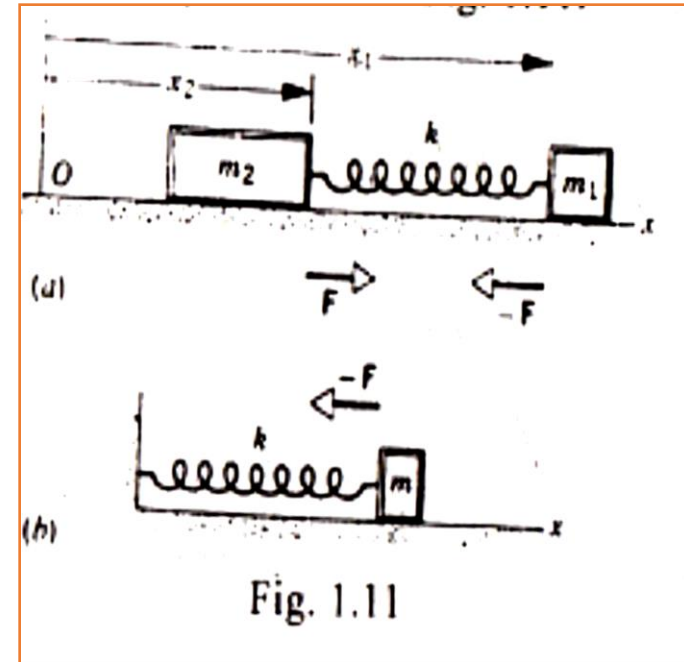


Two Body Oscillations

Newton's second law separately to the two particles

$$m_1 \frac{d^2 x_1}{dt^2} = -kx \quad \dots \quad (i)$$

$$m_2 \frac{d^2 x_2}{dt^2} = +kx \quad \dots \quad (ii)$$



By multiply the first of these equations by m_2 and the second by m_1 , we get

Two Body Oscillations

$$m_1 m_2 \frac{d^2 x_1}{dt^2} = -m_2 kx \dots \dots \text{(iii)}$$

$$m_1 m_2 \frac{d^2 x_2}{dt^2} = +m_1 kx \dots \dots \text{(iv)}$$

Subtracting equation (iv) from equation (iii), we can write

$$m_1 m_2 \frac{d^2 x_1}{dt^2} - m_1 m_2 \frac{d^2 x_2}{dt^2} = -m_2 kx - m_1 kx$$

Two Body Oscillations

$$m_1 m_2 \left(\frac{d^2 x_1}{dt^2} - \frac{d^2 x_2}{dt^2} \right) = -kx(m_1 + m_2)$$

$$\frac{m_1 m_2}{m_1 + m_2} \left(\frac{d^2 x_1}{dt^2} - \frac{d^2 x_2}{dt^2} \right) = -kx$$

$$\frac{m_1 m_2}{m_1 + m_2} \frac{d^2}{dt^2} (x_1 - x_2) = -kx$$

$$\frac{m_1 m_2}{m_1 + m_2} \frac{d^2}{dt^2} x = -kx$$

Two Body Oscillations

The quantity $\frac{m_1 m_2}{m_1 + m_2}$ has the dimension of mass and is known as the reduced mass of the system.

The equations of motion of two mutually interacting bodies can be reduced to a single equation describing the motion of one body in a reference frame centered in the other body. **The moving body then behaves as if its mass were the product of the two masses divided by their sum.** That quantity is called the reduced mass.

The unstretched length L of the spring is a constant. Hence the derivatives of $(x_1 - x_2)$ are the same as the derivatives of x .

$$\frac{d}{dt}(x_1 - x_2) = \frac{d}{dt}(x + L) = \frac{dx}{dt}$$

Two Body Oscillations

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

This is the differential equation of Simple Harmonic Motion of a single oscillating mass. m is the reduced mass of the system

The frequency of oscillation of the system is

$$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Mathematical Problems

Home Task: Example 1.16 to Example 1.21

Physics for Engineers (Part-1) -Dr. Gias Uddin Ahmed; pp: 53-59



Thank You
Thanks for your attention